



Project Title: (Live) Mobility KG for Vienna incl. various points-of-interest

Name: Nico Reiterer

Mode: ☒ 6 ECTS / ☐ 3 ECTS

Motivation. I wanted to work on something related to my surroundings. The City of Vienna provides quite rich open data and since I grew up here, I dug into what was possible with what I have easily available. The existing resources naturally led me to focus on typical points of interest and how to connect this to (live) data provided by Wiener Linien.¹

Problem. Sometimes it would be nice if somebody else planned a day or just a few hours of activities for you. Doing this yourself you have to (obviously) consider your interests, time constraints, travel times and such.

Solution. This project shall integrate public information on the City of Vienna with (live) Wiener Linien data into a unified KG. It will model transport links, points-of-interests, categories, spatial proximity (and temporal (live) data). Reasoning techniques shall be used to generate personalized activity plans and/or route suggestions based on user interests and (live) mobility conditions/constraints.

Steps.

- Data Collection & EDA: Analyse already collected datasets and explore how to combine them. Potentially expand or reduce scope.
- KG Modelling: Define a unified data model for transport nodes, POIs, categories, spatial relations, and temporal aspects.
- KG Creation: Map and ingest datasets using schema mapping and transformation techniques.
- Reasoning Layer: Implement logical reasoning for proximity, routing constraints, and do user preference filtering. (Optionally exploring KG embeddings.)
- Service Layer: Demonstrates how the KG could generate activity suggestions or route plans based on (live) data and user preferences.

Scope. Depending on the actual complexity, this project will focus on integrating Wiener Linien live data and at least 5 or so POI categories from publicly available data as a proof-of-concept. Emphasis is on KG schema creation, data integration and demonstrating basic KG reasoning services. If time and scope permit, GNNs and/or other advanced concepts will be explored.

I plan to focus on	LO5, LO6, LO7, LO9, LO11
I plan to show basic proficiency in	LO2, LO3, LO4, LO8, LO12
I plan not to include	LO1, LO10

¹ Note: Only after creating this one-pager I saw that the fallback project is quite similar to this...